**File System :**

It is the logical layer created on Partition. The purpose of creating FS on a partition is to make OS to perform Read & Write Operations.

- ➔ The Partition which is not formatted with FS called as RAW Device.
- ➔ The Partition which is formatted with FS called as block device.

**MKFS :**

It is the command used to create FS on partition. We can define the FS type as per our requirement. If we apply this command on any partition the complete data will be lost along with that it creates a new directory called lost + found.

Unfortunately if FS metadata is corrupted we cannot access the data from FS.

#mkfs.ext4 /dev/sdb1

#dumpe2fs -h /dev/sdb1 **(Displays the complete information of specific FS)**

**Mounting :**

**It is the process of attaching a FS to any one of the directory called as mounting. The purpose of mounting is to make FS accessible to users for performing read & write operations.**

#mkdir /sun

#mount /dev/sdb1 /sun

#df -h

**NOTE :** This mounting is temporary for the session, To make it permanent we need to update the FS information to /etc/fstab (Filesystem table)

**NOTE** : When we mount FS, The FS mounting information of all FS automatically updates to /etc/mtab file. If we unmount the FS information automatically removed from /etc/mtab file.

**To make FS permanent :**
#vi /etc/fstab

| Device to mount | Mount Point | FS type | Mount Options | FS Dump | FSCK pass |
|---|---|---|---|---|---|
| /dev/sdb1 | /sun | ext4 | defaults | 1 | 2 |

:wq!

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**FSDUMP :**

0 → Disabled
1 →Enabled
When this option is enabled at the time of crusial system failure core dump records the logs on this filesystem which may help for diagnostics purpose

**Difference Between EXT2 / EXT3 / EXT4 / XFS Filesystems :**

| Feature | EXT2 | EXT3 | EXT4 | XFS |
|---|---|---|---|---|
| **Journaling** | Not Available | Available | Available | Available |
| **Invented Year** | 1993 | 2001 | 2008 | 2012 |
| **Max File Size** | 2TB | 2TB | 16TB | 2EB |
| **Max FS Size** | 32TB | 32TB | 1EB | 8EB |
| **Max Directories** | 32K | 32K | 64K | 64K |

1024 byte = 1KB
1024 KB = 1MB
1024 MB = 1GB
1024 GB = 1TB
1024 TB = 1EB
1024 EB = 1PB
1024 PB = 1ZB

**FSC K pass:**

When the system is booting, Before the FS are mounted as per /etc/fstab file. The kernel checks this parameter. This parameter decides weather FSCK has to perform on particular FS or not.

If it is 0 → FSCK disabled on this FS
1 → recommended for Critical FS like ROOT
2 →Other user related FS

**UNMOUNTING :**

It is the process of detaching the FS from the ROOT dir. The most common reason we may go for unmounting FS is for either deleting partition or to perform some maintenance activities on partition.

**#umount /sun**

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**NOTE:** This unmounting is temporary for the session. To make it Permanent we need to remove this FS entry from /etc/fstab file.

**Unmounting Busy FS:**

We can unmount busy FS using 2 Methods :

1) Fuser command
2) Forcibly (Strictly Not Rec)

**Fuser :**

#fuser -cu /sun (Lists all the open process on /sun along with its owner)

#fuser -ck /sun (Kills all the open process on FS /sun)

#umount /sun

**Note : remove the entry from /etc/fstab.**

1) **Forcibly (Strictly Not Rec)**

#umount -f -l /sun

-f → Forcibly
-l →Makes FS first read only & then Unmount forcibly

**File system Check (FSCK)**

The purpose of FSCK is to bring the file systems from inconsistent to consistent state. FSCK should not apply on Mounted FS.

The most common reasons why GS are corrupted :

1) Metadata failure
2) Abnormal Shutdown

**We can run FSCK using 2 Modes :**

1) Interactive Mode (User Input is required to move forward)
2) Non-Interactive Mode (User input not required to move forward its takes default YES)

Running FSCK using Interactive MODE :
**#e2fsck /dev/sdb1**

Running FSCK using Non Interactive mode

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**#e2fsck -y /dev/sdb1**

-y --→ Non interactive

**Running FSCK with No changes ( Displays status of FS)**
**#e2fsck -n /dev/sdb1**

**-n** --→Running fsck with no changes

#e2fsck /dev/sdb1

#e2fsck -y /dev/sdb1

**NOTE :** When we run FSCK multiple times but still if the FS is not being recovered then it is the state where super block is corrupted

**Recovering super block using alternate super block :**

#dumpe2fs /dev/sdb1 |grep -I "superblock" (Displays Alternate superblock locations)

O/P

32659
86879
120589
178123
250741

Running fsck with the combination of alternate super block :

| | | | |
|---|---|---|---|
| #e2fsck | -b | <ABSBN> | <FS> |
| | | 32659 | /dev/sdb1 |
| | | 86879 | |
| | | 120589 | |

NOTE : We need to repeat this process with all the Alternate super block locations. For sure any one of the Alternate super block will match to recover the super block. Once it is done reboot the system.

**Most common Error codes in FSCk :**

0 - No errors
1 - File system errors corrected
2 - File system errors corrected, system should be rebooted

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4 - File system errors left uncorrected
8 - Operational error
16 - Usage or syntax error
32 - E2fsck canceled by user request
128 - Shared library error

**FS Architecture :**

1)Boot Block
2)Super Block
3)Alternate Super block OR Backup super block
4)Inode Block
5)Data Block

**Boot Block:**

It contains the initial boot strap program used to load the Operating system. The default location of Boot block is available in Sector ZERO.

**Super block:**

It contains complete information about your file system like :

1) Filesystem type
2) File system size
3) Block size and its utilisation
4) Inodes and its utilisation

**Note : The default location of super blocak is 16 to 31 sectors of the HDD.**

**Alternate Super block Or Backup Super block :**

Super block maintains identical copies of data in alternate super blocks. If unfortunately super block is corrupted we cannot able to acess the FS data until we recover the FS super block. We can recover the corrupted super block from alternate super block with the help of File system check utility.

There is no specific location for Alternate super block and it can be available in any part of the HDD and it depends on the size of the HDD.

**Command to display alternate super block:**

#dumpe2fs /dev/vdb1 -i "superblock"

**Inode Block :**

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It contains complete information about inodes.

**Data Block :** It contains complete information about Data

**NOTE:** These Inode blocks & Data Blocks are for kernel reference, These both does not have any specific location and can be available in part of the HDD.

**FS Journaling :**

A journaling filesystem keeps a journal or log of the changes that are being made to the filesystem during disk writing that can be used to rapidly reconstruct corruptions that may occur due to events such a system crash or power outage. The level of journaling performed by the file system can be configured to provide a number of levels of logging depending on your needs and performance requirements.

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